
DEPARTMENT OF INFORMATION TECHNOLOGY

Subject wise Common Question Bank (QB) for All Divisions

Class:	SY
Subject Name/Code:	Computer Networks (23UITL2303)
Subject Teacher Name(s):	1. Yugashree Bhadane 2. Dnyandeo Khemnar
Note: Only 60% Questions can be asked from this QB in CAE. Rest of 40% questions will be from out of this QB related to syllabus	

Unit-I	
Q. No.	Question Statement
1.	Define Network along with their types.
2.	Define Computer Network and write the applications that benefit from computer networks.
3.	Describe the role of a network device in a nutshell. Choose one specific network device (e.g., router, switch) and explain its function in more detail.
4.	Differentiate between Network Operating Systems (NOS) and network management software. Provide an example of each.
5.	Explain key benefits of using computer networks in an organization?
6.	Identify a security challenge associated with computer networks.
7.	Enlist and explain the seven layers of the OSI model.
8.	Explain TCP/IP model in detail.
9.	Compare and contrast the OSI model with the TCP/IP model.
10.	In your own words, define internetworking and explain its significance in today's world.
Unit-II	
Q. No.	Question Statement
1.	Briefly describe three types of transmission media used in networks (coaxial cable, twisted pair, fiber optic). Mention one advantage and one disadvantage of each type.
2.	Differentiate between Guided and Unguided Media.
3.	Differentiate briefly between analog and digital signals. Explain why digital signals are preferred for modern networks.
4.	List and briefly describe types network topologies with proper diagram also give advantage and disadvantage of each type.

5.	Explain how the choice of network topology can affect factors like scalability, performance, and troubleshooting in a network.
6.	What is the process of data encapsulation in the Data Link Layer, and why is it important?
7.	Explain the purpose of flow control protocols in the Data Link Layer. How do they prevent data overload on a network?
8.	Compare and contrast three flow control protocols: Stop-and-Wait, Go-Back-N, and Selective Repeat.
9.	Describe common error detection techniques used in the Data Link Layer. Briefly explain how each method identifies errors in data transmission.
10.	Explain how CSMA (Carrier Sense Multiple Access) is used to manage network access.